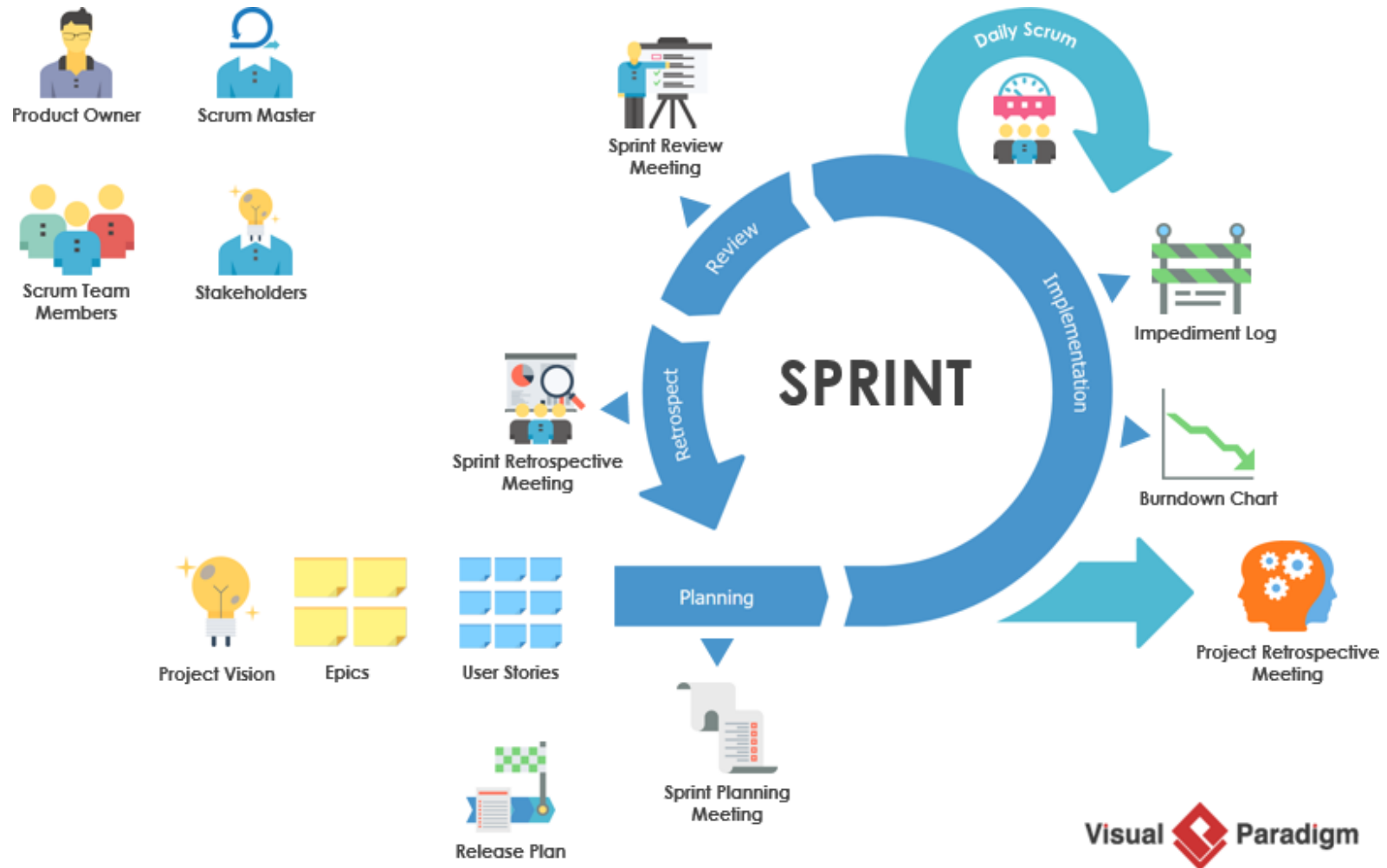


# Requirements Engineering

Praxisteil

# Projektvorgehen nach Scrum



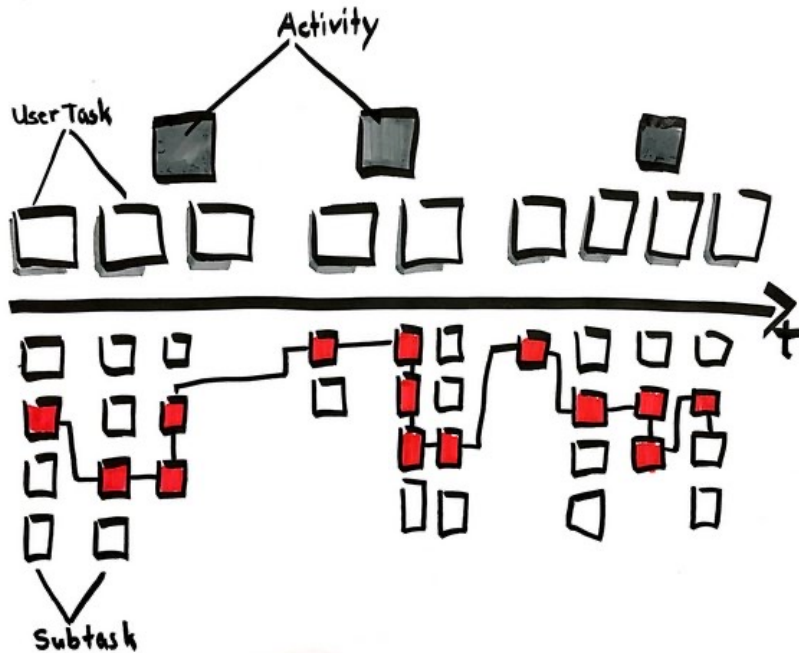
# Aufgabe: Projekt- / Produktvision

- **Formulieren Sie die Projekt-/Produktvision.**
- Erwartetes Ergebnis:
  - Tagline / Slogan für das Projekt (e.g., “Vorsprung durch Technik“)
  - One-Pager in PPT für die Beschreibung der Vision  
→ Stichwort: Elevator Pitch

- Lektüre: Übungsunterlage *Scrum mit User Stories - Was sind User Stories*
  
- Leitfragen beantworten:
  - a) Was ist ein Sprint in Scrum?
  - b) Was ist eine User Story?
  - c) Wie ist eine User Story aufgebaut? Gibt es Hinweise dazu, wie man eine User Story schreibt?
  - d) Was ist ein Epic? Was ist der Unterschied zwischen Epic und User Story?

# User Story Mapping

*Story Maps - Landkarten für das Product Backlog*



*„Story Mapping ist eine Technik, mit der ausgehend von der User Experience das Big Picture der Anforderungen übersichtlich dargestellt werden kann. Dabei bleibt stets der Kundenfokus erhalten.“*

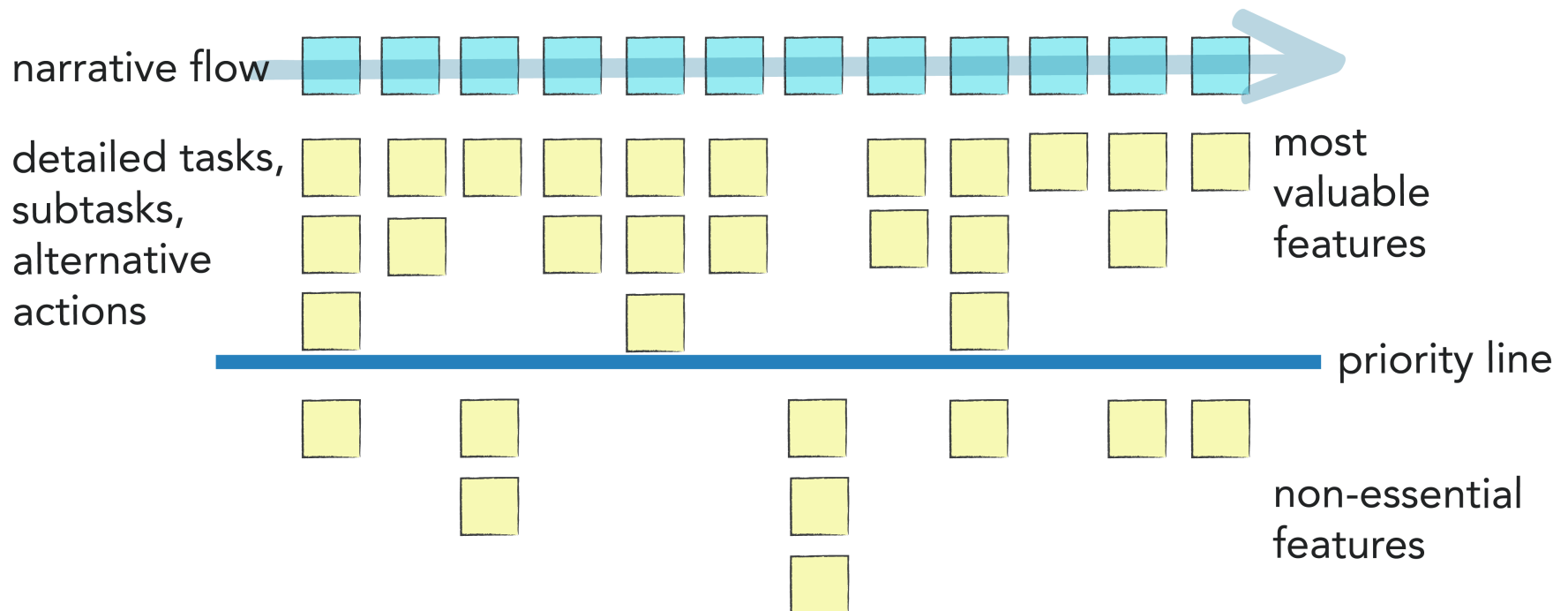
*Story Mapping beginnt mit den User Tasks; den Aufgaben, die aus Benutzersicht zu erledigen sind. Diese User Tasks ordnen wir horizontal sequenziell so an, dass sie eine nachvollziehbare Geschichte (Story) ergeben. Wir streben keine formale Modellierung an. Sicher werden User Tasks auch mal in einer anderen Reihenfolge ausgeführt oder bestimmte User Tasks entfallen in einer bestimmten Situation komplett“.*

Quelle: <https://www.it-agile.de/agiles-wissen/agiles-produktmanagement/story-mapping/>

# User Story Mapping

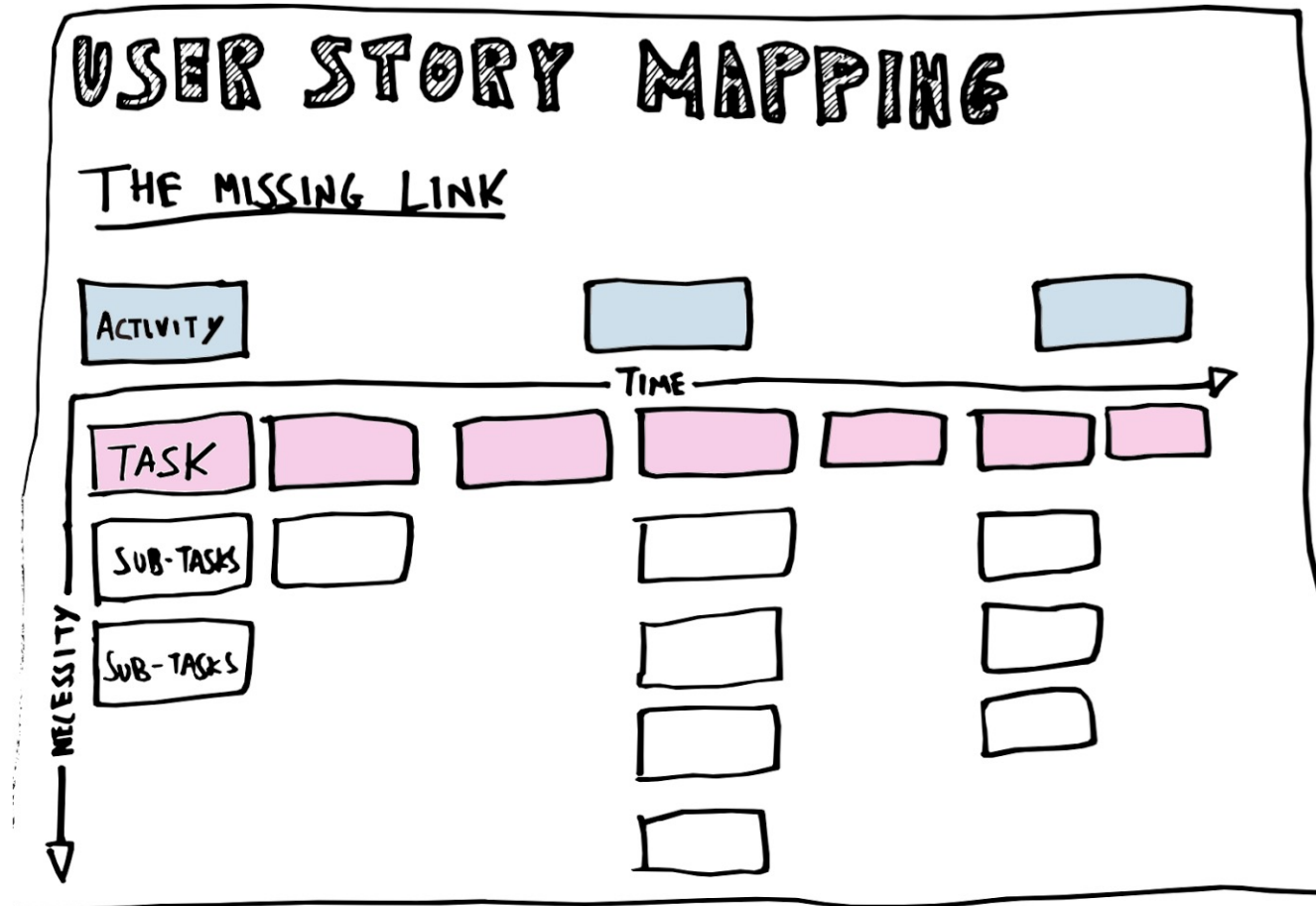
*Story Maps - Landkarten für das Product Backlog*

## User Story Map



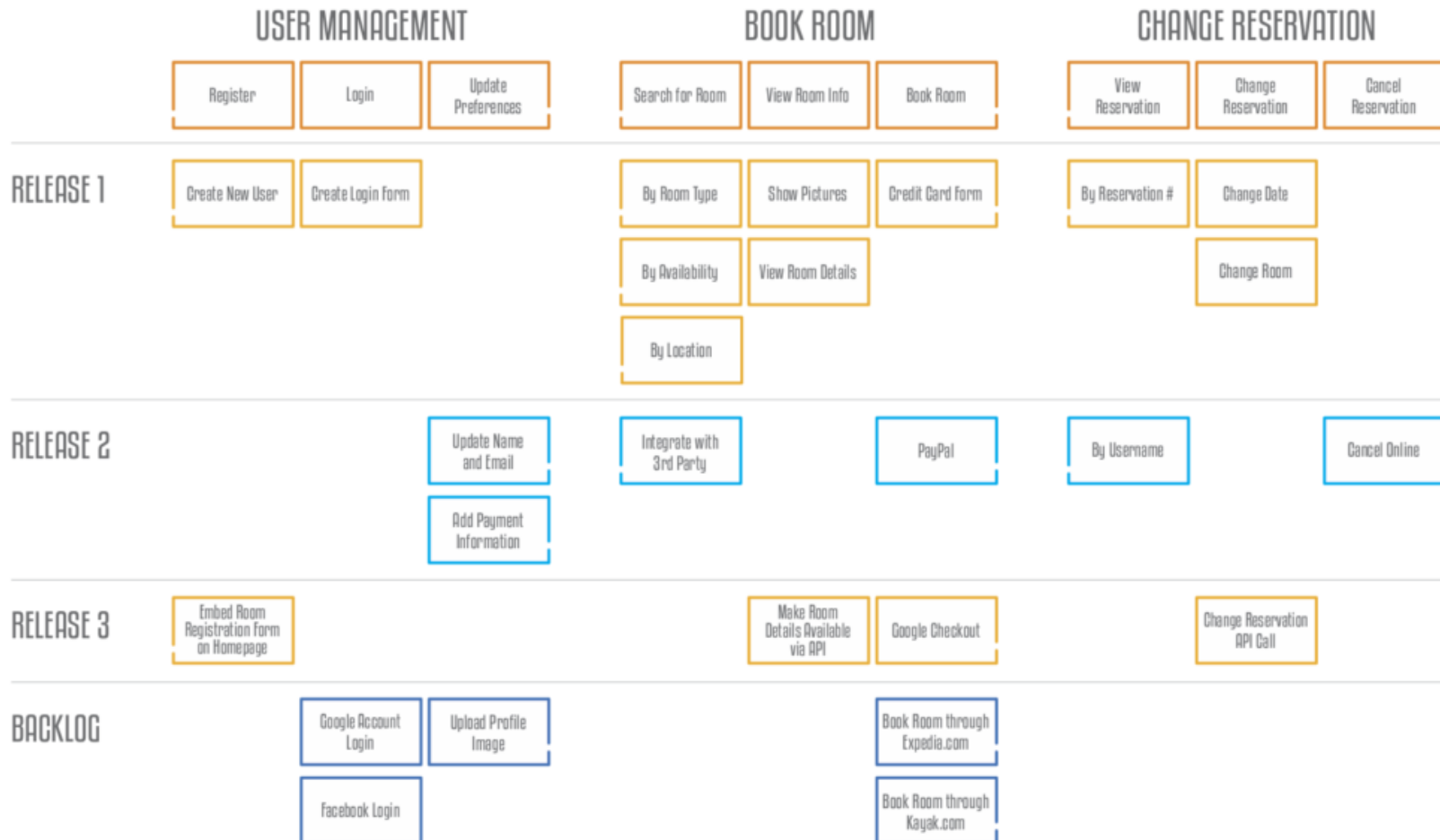
# User Story Mapping

Story Maps - Landkarten für das Product Backlog



# User Story Mapping

*Story Maps - Landkarten für das Product Backlog*



- **Erstellen Sie eine Story Map für Ihr Projekt / Ihre Software.**
- Erwartetes Ergebnis:
  - Story Map als PDF, PPT, XLS, o.ä.
  - Hinweis – Das Tool <https://storiesonboard.com/> ist recht komfortabel. Es gibt aber genügend andere Möglichkeiten, eine Story Map zu erstellen

## Definition:

- Personas veranschaulichen typische Vertreter einer Zielgruppe. Sie haben Erwartungen, Werte, Wünsche und Ziele und zeigen menschliche Verhaltensweisen. Eine Persona ist die Personifizierung bzw. der Prototyp einer Zielgruppe und hilft dabei, Annahmen über Kunden zu treffen.
- Als fiktive Anwender werden Personas mit konkreten Merkmalen wie Name, Foto, Wohnort, Lebenslauf, Familienstand, Alter, Einkommen, Ausbildung, Fähigkeiten, Einstellungen, Hobbies etc. beschrieben. Sie repräsentieren aber nicht den Durchschnitt einer Zielgruppe, sondern erscheinen als Personen, die Muster im Nutzungsverhalten erkennen lassen. Aus diesem Grund entwickeln Unternehmen oft auch mehrere Vertreter einer Zielgruppe.

Quelle: <https://t2informatik.de/wissen-kompakt/personas/>



Bildquelle: <https://www.cannypic.com/free-vector/free-personas-icons-vector-410369>

# Personas

*Personas veranschaulichen typische Vertreter einer Zielgruppe*



## Nerdy Nina

"The book is way better than the movie!"

#booklover  
#bookaddict  
#booknerdproblems

### DEMOGRAPHICS

Age: 25  
Location: Sao Paulo, Brazil  
Education: Software Engineer  
Job: Q/A at Indie Game Company  
Family: Lives with her boyfriend

### TECH

Internet  
Social Networks  
Messaging  
Games  
Online Shopping



### GOALS

- Discovering new books / authors to read
- Finding unique stories
- Cataloging book collection

### FRUSTRATIONS

- Keeping track of different series
- Forgetting a book launch date
- Finding space for more books

### READING HABITS

- Fast pace reader
- Never lends books
- Likes hardcovers and boxed collections
- Pre-order books to get them first
- Reads eBooks, but prefer physical copies
- Always finishes a book
- Loves binge reading and re-reading

### FAVORITE BOOKS



American Gods  
Neil Gaiman



Harry Potter  
J.K. Rowling



Ready Player One  
Ernest Cline

## Dozierende



### Eigenschaften:

- Prof. Dr. Doktor
- 40-60 Jahre
- muss überzeugt werden, engagiert
- Firmenzugehörigkeit: 10 Jahre  
hohes  
commitment

### Frustrations:

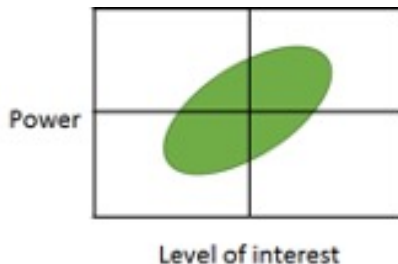
- Listen  
aushängen
- Unsicherheit,  
wie viele  
Studenten  
interessiert sind

### Spezifische Ziele:

- Einfachheit der System
- möglichst viele  
Studenten pro WPF
- Sicherheit über  
Zustandekommen  
eines Moduls

### Allgemeine Ziele:

- Wissen vermitteln
- Möglichst viele  
Studenten erreichen



### Nutzungsüberwindung:

- Bedienungsfreundlich
- Verbesserung des Ist-Zustandes → Abbau  
von Frustrationen
- Wenn Dozenten ihre Kurse nicht eintragen,  
blockieren sie die komplette Plattform

Tech savvy  
3-9/  
10

# Personas

*Personas veranschaulichen typische Vertreter einer Zielgruppe*

## ROMAN'S PERSONA TEMPLATE

 romanpichler

|  <b>PICTURE &amp; NAME</b>   |  <b>DETAILS</b>                                                                                                                                                                                                                              |  <b>GOAL</b>                                                    |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>What does the persona look like? What is its name? Choose a realistic and believable picture and name.</p> | <p>What are the persona's relevant characteristics and behaviours? For instance, demographics, such as age, gender, occupation, and income; psychographics, including lifestyle, social class, and personality; and behavioural attributes like usage patterns, attitudes, and brand loyalty. Only list relevant details.</p> | <p>What problem does the persona want to solve or which benefit does the character seek? Why would the persona want to use or buy the product?</p> |

www.romanpichler.com  
Template version 04/17

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# Personas

*Personas veranschaulichen typische Vertreter einer Zielgruppe*

## ROMAN'S PERSONA TEMPLATE

romanpichler

|  <b>PICTURE &amp; NAME</b>                                                                                                         |  <b>DETAILS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                |  <b>GOAL</b>                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>What does the persona look like? What is its name? Choose a realistic and believable picture and name.</p>  <p><b>Peter</b></p> | <p>What are the persona's relevant characteristics and behaviours? For instance, demographics, such as age, gender, occupation, and income; psychographics, including lifestyle, social class, and personality; and behavioural attributes like usage patterns, attitudes, and brand loyalty. Only list relevant details.</p> <p>Works as product manager for a mid-sized company.</p> <p>Is 35 years old, holds a marketing degree.</p> <p>Has got experience working as a product owner.</p> <p>Has managed mature products successfully.</p> | <p>What problem does the persona want to solve or which benefit does the character seek? Why would the persona want to use or buy the product?</p> <p>Needs to know how to determine the right strategy for a brand-new digital product.</p> |

www.romanpichler.com  
Template version 04/17

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# Personas

*Personas veranschaulichen typische Vertreter einer Zielgruppe*

## Mr/Mrs Müller



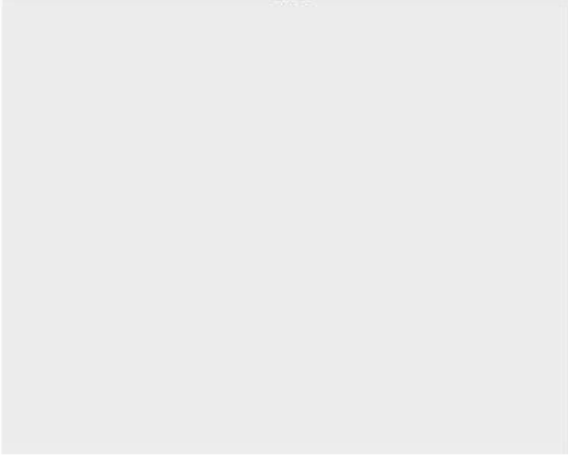
### DEMOGRAPHICS

AGE .....  
GENDER .....  
MARITAL STATUS .....  
INCOME .....  
LOCATION .....

### PROFESSIONAL BACKGROUND

EDUCATION .....  
COMPANY / JOB TITLE .....  
WORK EXPERIENCE .....

### BIO



### PERSONAL PREFERENCES

INTERESTS .....

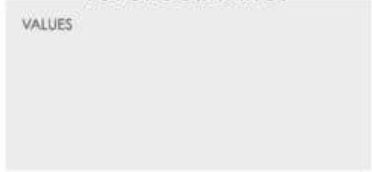


CHALLENGES & FRUSTRATIONS .....

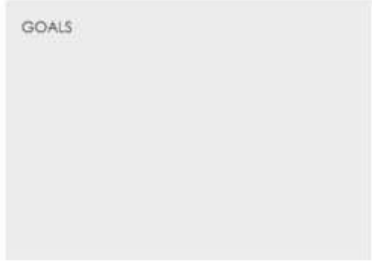


### PSYCHOGRAPHICS

VALUES .....



GOALS .....



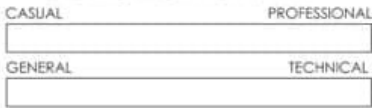
### PERSONALITY

EXTROVERT ..... INTROVERT .....  
THINKING ..... FEELING .....



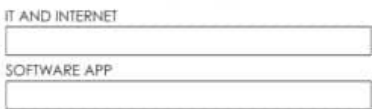
### COMMUNICATION STYLE

CASUAL ..... PROFESSIONAL .....  
GENERAL ..... TECHNICAL .....



### TECHNOLOGY

IT AND INTERNET .....  
SOFTWARE APP .....

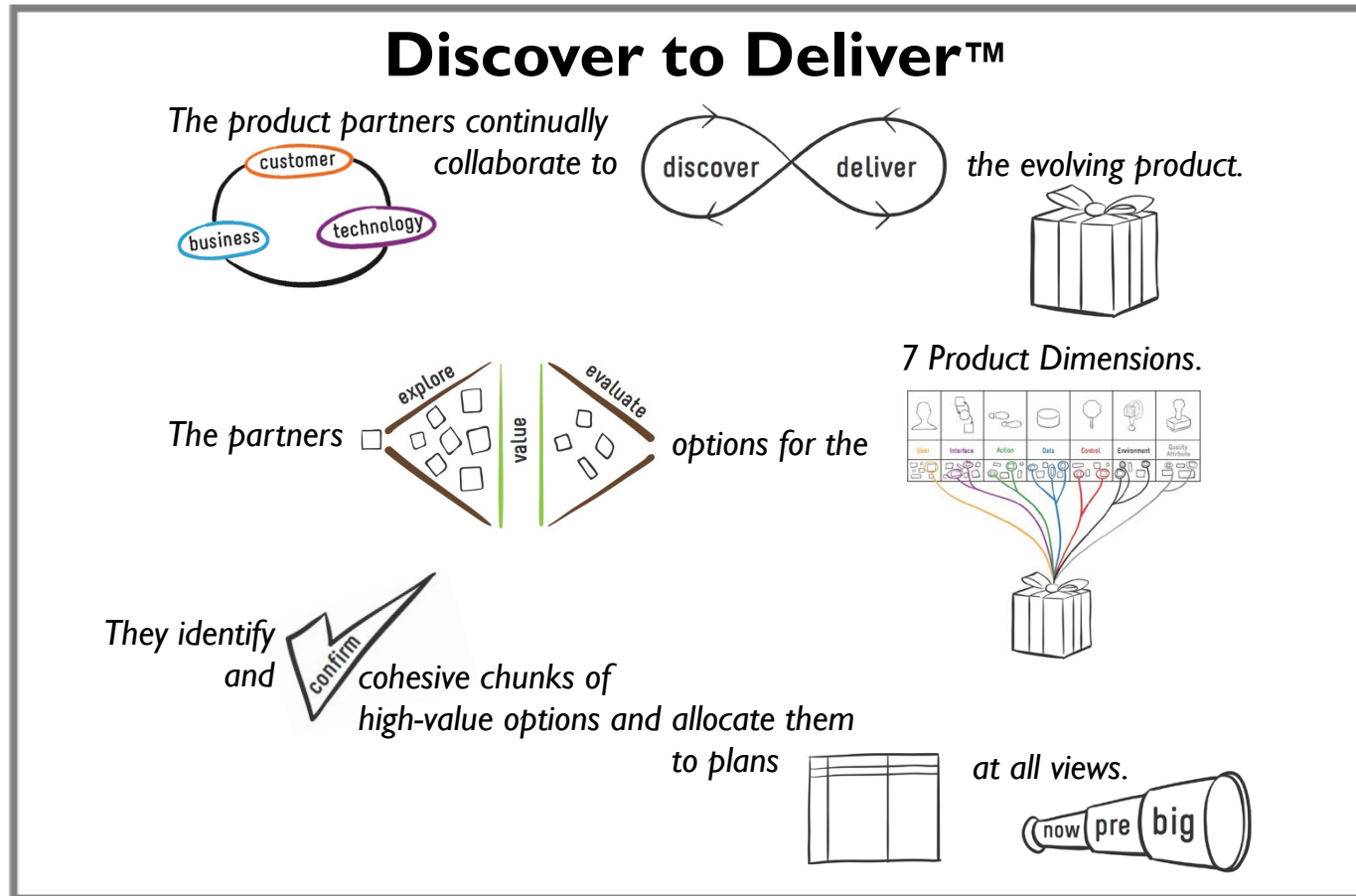


CLOU-MEDIA.DE

- **Erstellen Sie Personas für die zu entwickelnde Softwarelösung.**
- Erwartetes Ergebnis:
  - Persona als Grafik oder PPT/PDF
  - Hinweis – Sie können i.d.R. die Personas – d.h. die Rollen/Nutzer der Softwarelösung – aus der zuvor erstellten Story Map ableiten.

# Discover to Deliver

<http://www.discovertodeliver.com/>

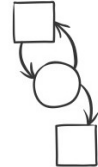
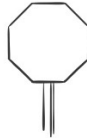


[View Key Terms Glossary](#)

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[www.ebgconsulting.com](http://www.ebgconsulting.com) | [www.DiscoverToDeliver.com](http://www.DiscoverToDeliver.com)



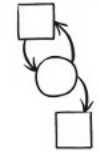
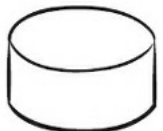
## 7 Product Dimensions

|                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |  |  |  |
| <b>User</b>                                                                       | <b>Interface</b>                                                                  | <b>Action</b>                                                                     | <b>Data</b>                                                                       | <b>Control</b>                                                                      | <b>Environment</b>                                                                  | <b>Quality Attribute</b>                                                            |
| Users interact with the product                                                   | The product connects to users, systems, and devices                               | The product provides capabilities for users                                       | The product includes a repository of data and useful information                  | The product enforces constraints                                                    | The product conforms to physical properties and technology platforms                | The product has certain properties that qualify its operation and development       |

# Discover to Deliver

<http://www.discovertodeliver.com/>

|                                                                                  |                                                                                   |                                                                                    |                                                                                    |                                                                                    |                                                                                    |                                                                                    |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
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## User

Users interact with the product.

Who values the product?

Who benefits from it?

Who uses it?



A product exists to serve its users. A user may be a person, another product, or a system that interacts with your product. A user may play many roles as they engage with the product. You explore and analyze the users to learn their roles and goals.

## User Examples

|           |                                        |
|-----------|----------------------------------------|
| Persona   | Charlie, a junior customer service rep |
| User Role | Service Scheduler                      |

# Discover to Deliver

<http://www.discovertodeliver.com/>

|                                                                                  |                                                                                   |                                                                                    |                                                                                    |                                                                                    |                                                                                    |                                                                                    |
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## User



- Stakeholder beschreiben
- Z.B. mittels Persona



**Nerdy Nina**  
"The book is way better than the movie!"

#booklover  
#bookaddict  
#booknerdproblems

**DEMOGRAPHICS**  
Age: 25  
Location: Sao Paulo, Brazil  
Education: Software Engineer  
Job: QA at Indie Game Company  
Family: Lives with her boyfriend

**TECH**  
Internet  
Social Networks  
Messaging  
Games  
Online Shopping

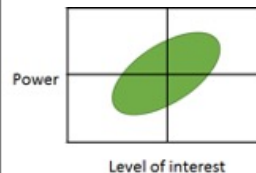
**GOALS**  
• Discovering new books / authors to read  
• Finding unique stories  
• Cataloging book collection

**FRUSTRATIONS**  
• Keeping track of different series  
• Forgetting a book launch date  
• Finding space for more books

**READING HABITS**  
• Fast pace reader  
• Never lends books  
• Likes hardcovers and boxed collections  
• Pre-order books to get them first

**FAVORITE BOOKS**  


## Dozierende



### Eigenschaften:

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- muss überzeugt werden, engagiert
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hohes  
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3-9/  
10

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# Discover to Deliver

<http://www.discovertodeliver.com/>

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## Interface

A product connects to users, systems, and devices.

What interfaces do you need to build for the users to interact with the product?

How does the product receive data?

How does the product send data?



An interface is the mechanism that users or other systems employ to exchange data with the product. You may choose to prototype new or changing interfaces as a means to discover and analyze product options.

## Interface Examples

|                            |                                                                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| User interface             | Customer provides service request details.<br>Customer receives a confirmation.<br>Customer service rep calls a customer to follow up on a completed job. |
| Hardware device interface  | Cleaner's handheld device sends the customer's job completion signature to SK.                                                                            |
| System-to-system interface | SK's Inventory system sends an order to the Wholesale Distributor.                                                                                        |

# Discover to Deliver

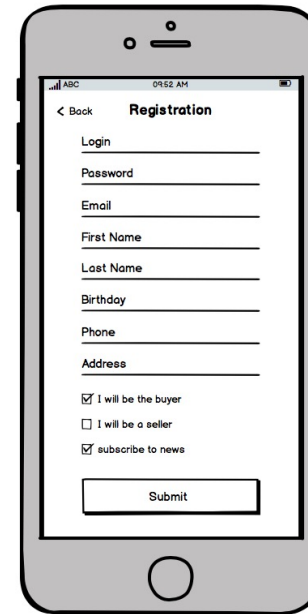
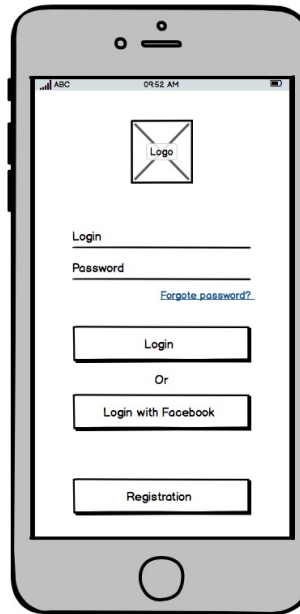
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|                                                                                  |                                                                                   |                                                                                    |                                                                                    |                                                                                    |                                                                                    |                                                                                    |
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## Interface



- GUI-Design beschreiben
- Schnittstellen (SST) zu Nachbarsystemen (NS) als Kontextdiagramm oder ggf. als Datenflussdiagramm
- Art der Anbindung von NS-SST darlegen

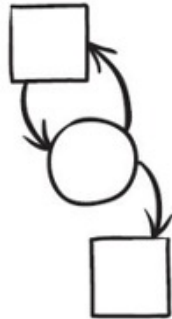


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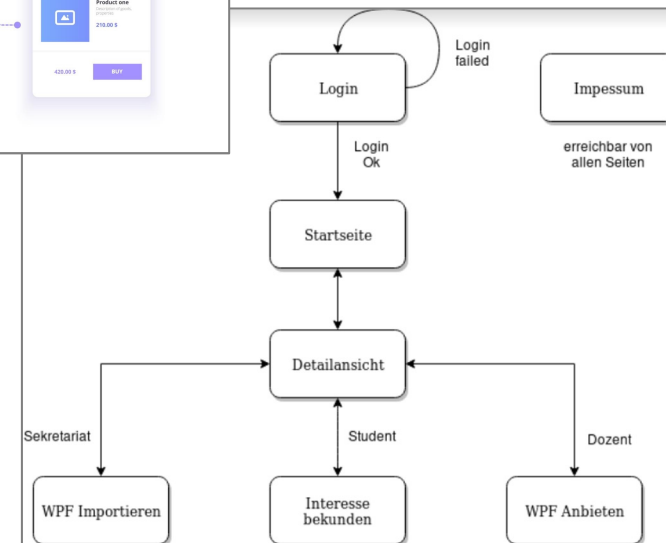
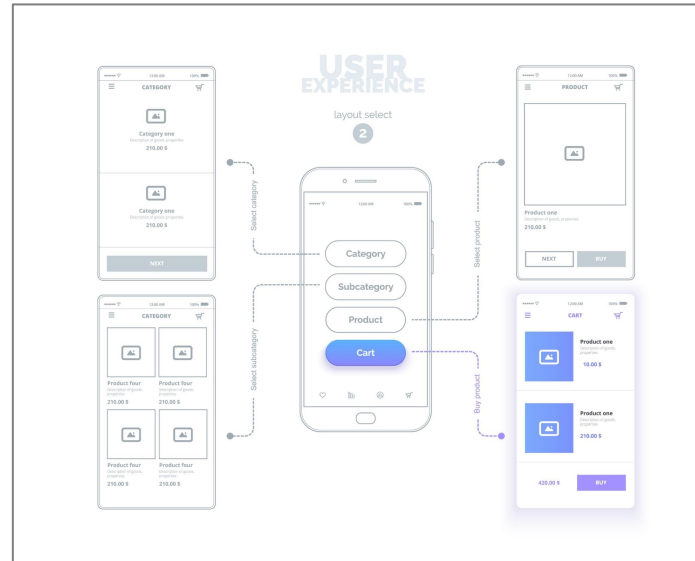
<http://www.discoverto deliver.com/>

|                                                                                  |                                                                                   |                                                                                    |                                                                                    |                                                                                    |                                                                                    |                                                                                    |
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## Interface



- Spätestens nach Beschreibung mehrerer GUI-Dialoge ergibt sich eine *Dialoglandkarte*



# Discover to Deliver

<http://www.discovertodeliver.com/>

|                                                                                  |                                                                                   |                                                                                    |                                                                                    |                                                                                    |                                                                                    |                                                                                    |
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## Action

The product provides capabilities for users.



How do actions provide business value?

How is an action triggered?

How does the product respond?

How are actions sequenced?

How do actions impact data?

The Actions Dimension of a product represents activities that achieve results that the users need. An event triggers an action to execute the product's planned responses (which are also actions). The event may be initiated by users, by a predefined time, or by a predecessor action. An action may enforce controls (e.g., policies, rules). Actions transform data into usable information for users.

You elaborate and decompose high-value actions to an atomic level that enables you to unambiguously develop and test the action. You allocate actions that will be automated to a software solution. Non-automated actions must be integrated with the software and supported by it.

## Action Examples

|                  |                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------|
| Event            | Customer contacts SK to arrange for a service.                                                          |
| Business Process | Schedule a Job                                                                                          |
| User Story       | As a customer, I need to schedule services at a time that is convenient for me so that...               |
| Scenario         | Roger Smith needs to schedule window cleaning services at his residential property on Saturday morning. |

# Discover to Deliver

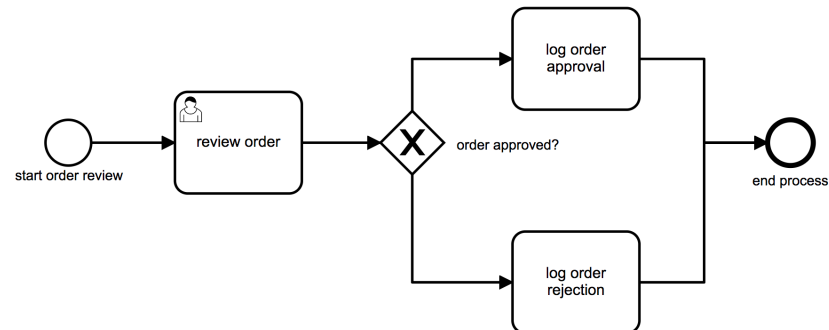
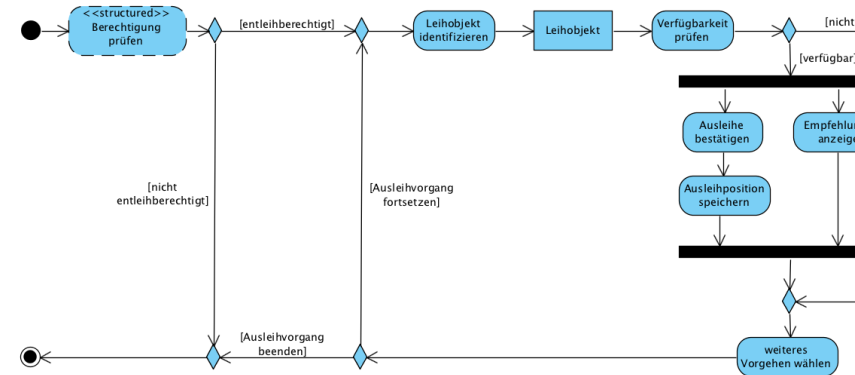
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|  |  |  |  |  |  |  |
| User                                                                             | Interface                                                                         | Action                                                                             | Data                                                                               | Control                                                                            | Environment                                                                        | Quality Attribute                                                                  |
| Users interact with the product                                                  | The product connects to users, systems, and devices                               | The product provides capabilities for users                                        | The product includes a repository of data and useful information                   | The product enforces constraints                                                   | The product conforms to physical properties and technology platforms               | The product has certain properties that qualify its operation and development      |

## Action



- Beschreibung der Abläufe, die in der User Story abgebildet werden.
- Was kann man in der Anwendung wann machen?
- Was macht das System automatisch?
- Wie bettet sich die Nutzung der Anwendung in den Gesamt-Prozesskontext ein?
- Etc. pp.



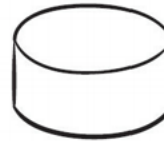
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## Data

A product includes a repository of data and useful information.



What data do users need from the product?

What data does the product receive? From where?

What data must the product store? How do you know whether the data is valid?

For how long is the data valid? What is the context in which the data has meaning?

The Data Dimension of a product represents data (raw facts) and information (interpreted data) needed to support the product's users. Data and information are essential for making business decisions. Actions transform data into a useful context for users. The sources of data include external systems and users.

You elaborate and decompose high-value data into elementary *data attributes*. For data that must be represented and accounted for in the software, you use data models showing all product data at rest, organized into groupings and relationships. As the product grows in functionality, the data to be stored also expands.

## Data Examples

|                |                |
|----------------|----------------|
| Data Domain    | Financial      |
| Data Entity    | Payment        |
| Data Attribute | Payment amount |

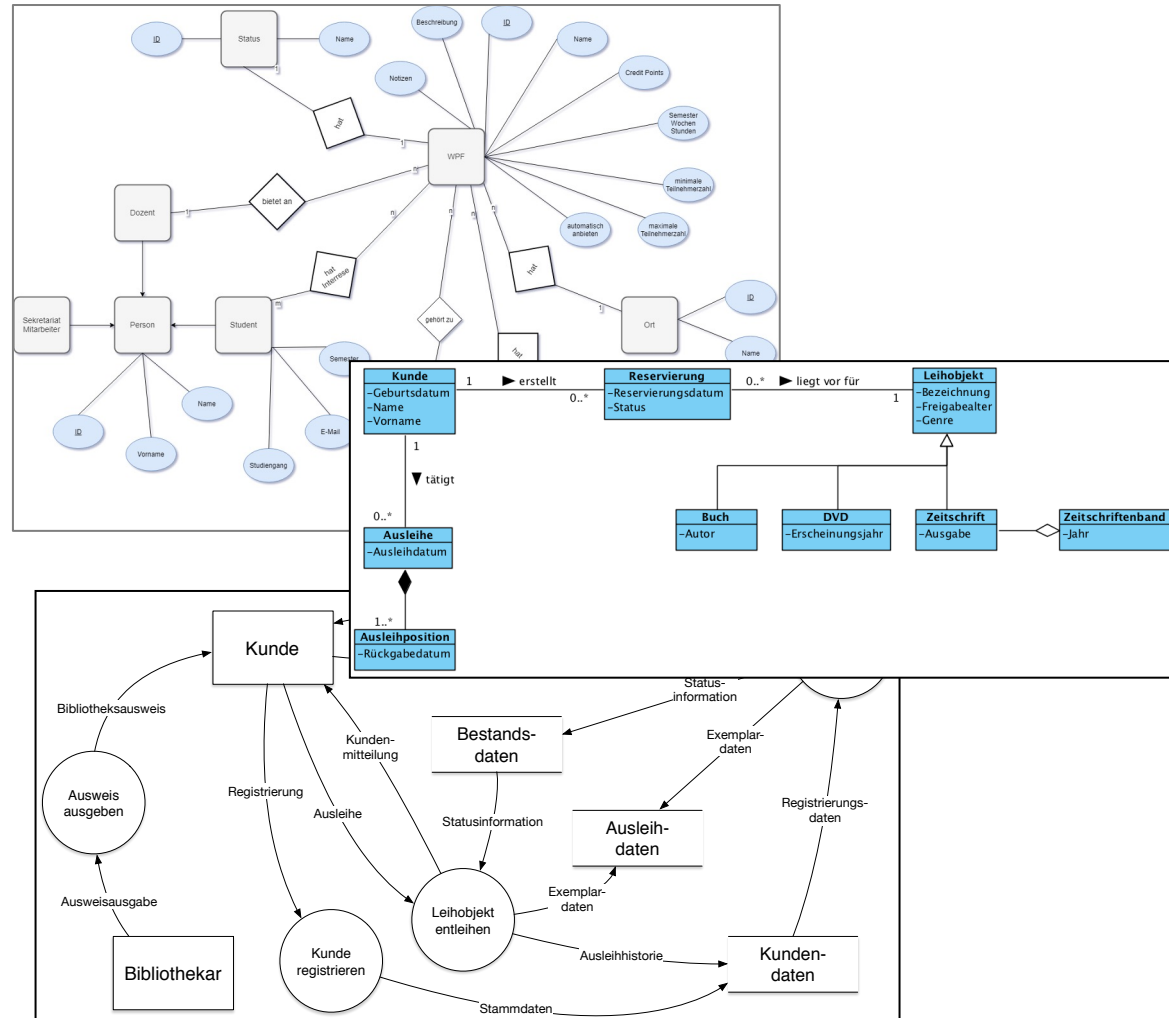
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|  |  |  |  |  |  |  |
| <b>User</b>                                                                      | <b>Interface</b>                                                                  | <b>Action</b>                                                                      | <b>Data</b>                                                                        | <b>Control</b>                                                                     | <b>Environment</b>                                                                 | <b>Quality Attribute</b>                                                           |
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## Data

- Wie sieht das Gesamt-Datenmodell aus?
- Welcher Ausschnitt aus dem Gesamt-Datenmodell ist für die User Story relevant?
- Welche Daten werden wo angelegt, angezeigt, geändert, gelöscht?
- Sind alle Daten im System, oder werden Daten von Services / Nachbarsystemen bezogen und diesen zur Verfügung gestellt?



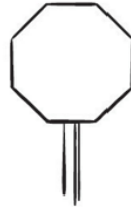
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## Control

A product enforces constraints.



Why are you motivated to conform to policies and regulations?

What are the risks if you do not comply with control needs?

What violations must be detected?

What is the jurisdiction of a given policy, regulation, or rule?

What is the enforcement level for the rule?

The Control Dimension of a product represents guidelines or constraints on the conduct of the business. They provide a basis for decision making. A system must be able to evaluate such controls and use them to direct appropriate action. Whether the controls originate from the business itself or from external sources (e.g., regulatory agencies), you must articulate controls in a manner that can be validated. Typically, you express controls as clearly defined statements or acceptance tests.

Controls exist regardless of their enforcement mechanism—whether they are enforced in software, manually through a business process, or both. When necessary for compliance, controls must be traceable from code back to the originating source.

You can organize complex controls in *decision tables* or *decision trees*. The team determines whether a control will be enforced via software or manually as part of human work, depending on value, risks, and dependencies.

## Control Examples

|                 |                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------|
| Business Policy | A customer who has not paid for prior services may be prevented from scheduling new services.                      |
| Business Rule   | An individual customer whose total delinquent payment amount is greater than \$250 must not schedule new services. |

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## Environment

### A product conforms to physical properties and technology platforms.



Where will the product be used? From what physical locations?

What are the product's physical properties?

What technologies and physical means will be used to install, configure, access, revise, license, protect, adapt, or dispose of the product?

Does the product need to conform to certain technology standards?

How do development and operational environment options affect quality attributes?

The Environment Dimension of the product includes the operational and development environment.

The **operational environment** includes the in-use product's physical properties and its technology platform. The operational environment is specific to a location and possibly to specific users and product actions. This environment has physical properties that need to be understood, such as the place and space where the product will be operated, the product's external and internal composition, and mechanisms for accessing the product.

The **development environment** governs how the product will be architected, developed, tested, deployed, and released into service. The development environment is composed of the hardware, software, and standards that make up its technology platform.

Both the operational and the development environments use a **technology platform**.

The technology platform includes hardware components, software components, and the standards used to develop and operate the product. Technology platform hardware includes physical equipment, media, and devices that contain software. Technology platform software is the set of components that manage the plumbing underlying the product. Standards provide common protocols for software and hardware development, deployment, and operation.

### Operational Environment Examples

| User | Location Physical | Properties                    |
|------|-------------------|-------------------------------|
| CSR  | SK office         | Noisy, congested office space |

### Development Environment Examples

| Software        | Hardware          | Standards                   |
|-----------------|-------------------|-----------------------------|
| GreatBrowser3.2 | Cassidy3500Router | DistributedDataStandard 4.3 |

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## Quality Attribute

A product has certain properties that qualify its operation and development.



How will you know whether the quality attributes satisfy the partners' needs?

How will you test and measure the quality attributes?

Do quality attributes apply to all or only a subset of the product options?

Can the quality attributes be implemented incrementally?

What degree of specificity do you need?

The quality attributes are predefined levels of service for the product's functionality and interfaces. Quality attributes describe operation and development properties for the product's User, Interface, Action, and Data dimensions. For example, a user story might be qualified by multiple quality attributes, such as performance, security, and usability. A quality attribute might be specific to one or more users, interfaces, actions, or data entities. Some quality attributes—for example, audit, authentication, localization, reliability, and security—may apply to more than one dimension.

## Quality Attribute Examples

|                |                                                                                                                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Availability   | The SK scheduling application will be available 95% of the time Monday through Friday from 8:00 a.m. to 5:00 p.m. Eastern.<br>The SK scheduling application will be available 95% of the time on Saturday from 8:00 a.m. to 2 p.m. Eastern. |
| Installability | An untrained user will be able to install upgrades to the SK application within 10 minutes without accessing any printed documentation.                                                                                                     |

# Aufgabe: Beschreibung einer User Story nach D2D-Ansatz

- Wählen Sie eine User Story aus Ihrer Story Map aus und beschreiben Sie diese detailliert(er) nach D2D-Ansatz.
- Erwartetes Ergebnis:
  - Betrachtung der 7 Dimensionen nach D2D in Bezug auf die gewählte User Story.
  - Dokumentation der spezifizierenden Inhalte für jede der 7 Dimensionen.
  - Hinweis: Sollten Sie der Meinung sein, dass für die von Ihnen gewählte User Story die Betrachtung einer bestimmten Dimension nicht zielführend ist oder es dazu keine Aussagen zu treffen gibt, dann dokumentieren Sie dies entsprechend.

## Gruppenpräsentation (alle Teilnehmer:innen zusammen):

- Präsentation Ihrer Ergebnisse vor dem Auftraggeber (AG). Es geht v.a. um Inhalt, aber natürlich auch um eine ansprechende Präsentation, die dem AG ein gutes Gefühl gibt, mit dem richtigen Auftragnehmer (AN) zu arbeiten.

## Inhalte (Vorschlag):

- Überblick und Produktvision
- System- und Kontextabgrenzung
- Stakeholderüberblick (e.g., Personas)
- Story Map inkl. Releaseplanung für drei Releases mit möglichst realistischer Priorisierung
- Gesamt-Datenmodell (System-eigene Daten und Daten via Schnittstelle zu Nachbarsystemen)
- Dialoglandkarte / Sitemap
- Detaillierte Vorstellung von 3 User Stories (Dozent:in bietet Termin mit Slots an, Interessent:in bucht Terminslot, Interessent:in storniert Terminslot). Dokumentiert nach D2D und den 7 Dimensionen.
- Testfälle (nach Given – When – Then oder anderem Schema)